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MieSimulatorGUI: A user-friendly tool to compute and visualize light scattering by spherical dielectric particles

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Summary

Mie theory is a mathematical framework derived from Maxwell's equations that models electromagnetic scattering by spherical dielectric particles. Its predictions are essential across various scientific and engineering disciplines, including biomedical optics, atmospheric optics, particle characterization, nanofluids, computer graphics, and remote sensing. To bridge the gap for researchers who require Mie simulations but lack specialized programming expertise, we developed `MieSimulatorGUI`. By integrating a high-performance C/C++ computational engine with the Qt framework, this user-friendly cross-platform tool calculates key optical properties such as scattering coefficients, cross-sections, angular scattering distributions, and scattering asymmetry for both monodisperse and polydisperse particle distributions. The graphical user interface (GUI) features six interactive panels that allow users to specify input parameters, visualize distributions, and analyze scattering phenomena. Furthermore, it enables the fitting of spectrally varying reduced scattering coefficient parameters, a feature particularly valuable in fields like tissue optics. `MieSimulatorGUI` is open source, hosted on GitHub and distributed under the MIT license; it is available via the project's download page.

Statement of need

Mie theory, derived from Maxwell's equations, provides a comprehensive framework for modeling electromagnetic scattering by spherical particles (Mie, 1908). This theory is invaluable across diverse fields, ranging from nanomaterials and biomedical optics to atmospheric science and astronomy (Bhandari et al., 2011; Chalut et al., 2008; Goody & Yung, 1989; Horvath, 2009; Saidi et al., 1995; Wang et al., 2005). Despite its broad applicability, the theory's reliance on complex mathematical constructs, such as infinite series and special functions (Bohren & Huffman, 1983; Hulst, 1957; Majic & Le Ru, 2020; W. J. Wiscombe, 1980), demands advanced computational implementation.

While numerous Mie simulation packages are available (many of which are listed on SCATTPORT.org and Wikipedia), they generally fall into two categories: older, established codes focusing on computational efficiency (Bohren & Huffman, 1983; W. J. Wiscombe, 1980), and newer, object-oriented libraries typically hosted on version-control platforms (Giannakopoulos, Ilias, 2018; Prahl, 2023; Sivry-Houle et al., 2023; Sumlin et al., 2018). Although both categories provide robust computational engines, they usually demand significant programming proficiency. This requirement creates a barrier for experimentalists, clinical scientists, and educators who need these analytical capabilities but may lack the specialized coding expertise to integrate such libraries into their workflows.

`MieSimulatorGUI` bridges this gap by providing an intuitive, cross-platform desktop application

that computes and fits scattering parameters for monodisperse and polydisperse distributions without requiring any coding. Unlike standard implementations, it supports heterogeneous polydispersity, allowing users to assign bin-specific complex refractive indices via custom data inputs, a feature often absent in simplified GUI tools. The tool facilitates high-impact use cases such as biomedical optics (Jacques, 2013; Mourant et al., 1997; Wang et al., 2005) and atmospheric research (Seinfeld & Pandis, 1998; Teri et al., 2022), where users can define complex configurations and directly fit spectrally varying reduced scattering coefficients. By integrating a powerful C/C++ computational engine with intuitive Qt interface, `MieSimulatorGUI` offers accessible yet powerful Mie theory computations, facilitating both streamlined research analysis and interactive pedagogical demonstrations.

Main Features

Built on the BHMIE (Bohren & Huffman, 1983) and Wiscombe (Warren J. Wiscombe, 1979) frameworks, `MieSimulatorGUI` calculates spectral optical properties, including scattering coefficients, cross-sections, scattering amplitude matrix entries (S_1, S_2), phase functions, and scattering asymmetry. The tool supports both monodisperse and polydisperse particle distributions (Gélébart et al., 1996) and facilitates parameter estimation by fitting reduced scattering coefficient curves, a technique of significant value in tissue optics (Jacques, 2013). To ensure computational accuracy and GUI stability, the software includes an automated test suite integrated via GitHub Actions. Its high-performance C++ engine enables near-instantaneous computation and plotting, providing real-time visual updates across the spectral range. Comprehensive documentation, including cross-platform installation guides (Windows, Linux, and macOS), dependency specifications, command-line test execution and several examples, is available on the Mie Simulator GUI Wiki page.

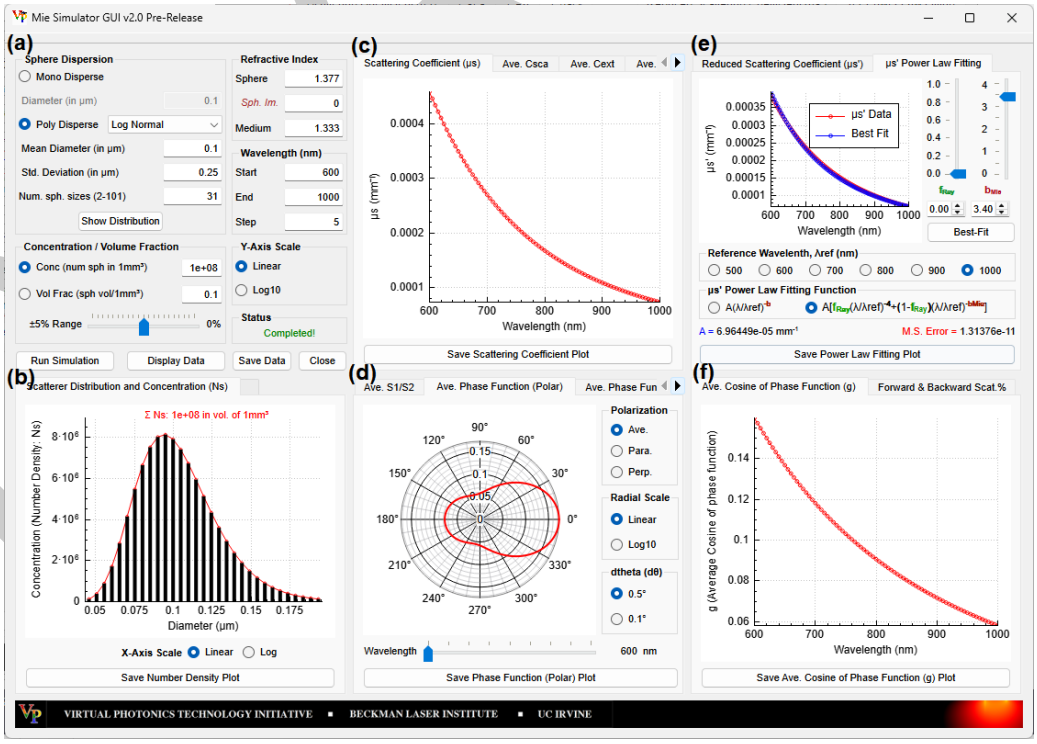

Figure 1: Six interactive panels of `MieSimulatorGUI`: (a) Input selection, (b) Particle size distribution, (c) Scattering coefficient, (d) Phase function, (e) Reduced Scattering, and (f) Scattering Asymmetry (Anisotropy)

Design and Functionality

The tool is distributed as portable binaries for Windows, macOS, and Linux. For local compilation, the project utilizes `qmake`, with dependencies (`Qt6` and `QCustomPlot`) managed via an automated build script. The Qt GUI contains six interactive panels (Figure 1).

Input Selection Panel

This panel enables the user to define inputs for Mie simulations at either a single wavelength or across a spectral range. The distribution of spheres is described using sphere concentration (`Conc`) (spheres/mm^3) or volume fraction (`Vol Frac`). The volume fraction, the proportion of space occupied by spheres, is determined by the product of the sphere concentration in 1 mm^3 and the individual sphere volume. `MieSimulatorGUI` utilizes the independent scattering approximation, a framework valid for dilute suspensions where particles are sufficiently separated to ignore coherent interactions (Hulst, 1957; Schmitt & Kumar, 1998). Since the validity of this approximation depends on the inter-particle spacing, size parameter, and relative refractive index, its accuracy diminishes as the volume fraction increases (Galy et al., 2020; Ivezić & Mengüç, 1996). Consequently, the tool is best suited for dilute systems; the results obtained for concentrated regimes may deviate from physical reality and should be interpreted with caution.

To maintain numerical stability in the BHMIE algorithm and ensure UI responsiveness, sphere diameters are restricted to a range of 0.1 nm to 300 µm and wavelengths are limited to 50 nm – 3000 nm. These ranges cover the primary biomedical and atmospheric spectral windows. For absorbing spheres, the complex refractive index (m_{sphere}) is defined as m_{real} – j m_{imag}, where m_{real} and m_{imag} represent real and imaginary components, respectively (Hulst, 1957; Warren J. Wiscombe, 1979).

The tool provides options for either monodisperse (uniform-sized) or polydisperse (variable-sized) particle distributions. Monodisperse distributions restrict analysis to spheres with uniform size and refractive index. In contrast, polydisperse distributions enable simulations of spheres with diverse attributes and support three size distribution models: 1. Log-normal, 2. Gaussian, and 3. user-defined (custom). The user-defined option allows for the specification of different refractive indices for different spheres, as demonstrated in the examples provided in the `CustomDataSamples` folder.

Number Density Panel

This panel graphically presents the number density of spheres N_s [#/ mm^3] used in the simulation. The subsequent tab displays the `Size Parameter` defined as $2\pi R n_{med}/\lambda_{vacuum}$ (Bohren & Huffman, 1983), where R [µm] denotes the particle radius, n_{med} the medium's refractive index, and λ_{vacuum} [µm] is the wavelength in vacuum.

Scattering Coefficient Panel

The Mie calculations provide three important efficiency factors: the scattering efficiency (Q_{sca}), the extinction efficiency (Q_{ext}), and the backscattering efficiency (Q_{back}) (see Mie Scattering Efficiencies). These dimensionless quantities combined with the particle's cross section area (πR^2) yield the corresponding scattering cross section C_{sca} [/mm^2], extinction cross section C_{ext} [/mm^2] and backscattering cross section C_{back} [/mm^2] (Bohren & Huffman, 1983; Hulst, 1957). The calculated cross sections are displayed across three separate tabs. For monodisperse distribution, the scattering coefficient (μ_s) is simply the product of the scattering cross-section C_{sca} and the number density N_s. For polydisperse distributions, the scattering coefficient is computed via a discrete summation of the cross-sections across individual particle size bins, as detailed by (Schmitt & Kumar, 1998).

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Phase Function Panel

The phase function represents the angular distribution of scattered light. The calculated phase function results are displayed in this panel using both polar and linear plots. These plots are derived from the complex amplitude scattering matrix elements, S_1, and S_2, which describe the transformation of incident electromagnetic field to far-field scattered field (Bohren & Huffman, 1983; Hulst, 1957). The wavelength slider allows the user to visualize the phase function or S_1 and S_2 data at a specific wavelength.

Scattering Asymmetry Panel

The scattering asymmetry (Anisotropy) panel displays the directional properties of the scattering phase function. The first tab presents the average cosine of the phase function (g), which quantifies the average scattering angle which indicates the prevalence of forward ($g > 0$) vs backward scattering ($g < 0$). The second tab provides the integrated forward and backward scattering fractions, offering a detailed analysis of the angular scattering distribution.

Reduced Scattering Panel

This panel shows the reduced scattering coefficient (μ_s'), calculated as the product of the scattering coefficient (μ_s) and $(1-g)$ (Jacques, 2013). This parameter is crucial in various fields, particularly in biomedical optics, because it allows for the non-invasive quantification of tissue properties. Users can use `μs' Power Law Fitting` tab to compute the fitting parameters that provide a simplified functional form for the wavelength dependence of μ_s' as described in (Jacques, 2013).

Example Application: Scattering of Intralipid Phantoms

To demonstrate the tool's scientific utility, consider the characterization of Intralipid 20%, a common tissue phantom in biomedical optics (Di Ninni et al., 2011). Users can define the medium by inputting polydisperse particle distribution parameters using a `Log Normal` distribution. For Intralipid 20% (w/w), which corresponds to a volume fraction (`Vol Frac`) of 0.227 (Aernouts et al., 2013), users may assume a mean particle diameter of 0.22 μm and a standard deviation of 0.4 μm, following Raju (2017) (Raju & Unni, 2017). After setting the `Num. sph. sizes` field to 101, users can specify the refractive index of the soybean oil droplets as 1.47 and the surrounding medium as 1.33. Upon executing the simulation across the 600–1000 nm spectral range, `MieSimulatorGUI` calculates μ_s, μ_s' and g. These values are directly comparable to experimental data (Aernouts et al., 2013) and can be exported as text data files for further analysis.

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